Fiscal Year:	FY 2022	**Task Last Updated:**	FY 01/05/2023
PI Name:	Delp, Michael Ph.D.		
Project Title:	Radiation, Simulated Weightlessness and Countermeasures: Effects on Cerebral and Coronary Vascular Function and Structure		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	08/01/2019	**End Date:**	07/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	End date changed to 07/31/2023 per NSSC (Ed., 8/25/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Xiao Wen M.D. (Loma Linda University) Muller-Delp, Judy Ph.D. (Florida State University) Shirazi-Fard, Yasaman Ph.D. (NASA Ames Research Center) Ledermann, Thomas Ph.D. (Florida State University)		
Grant/Contract No.:	80NSSC19K1322		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is one of two proposals selected for funding in response to solicitation NNJ16ZSA001N-SRHHC “Topic 6: Combined Effects of Simulated Weightlessness and Space Radiation on Cardiovascular Structure and Function,” and then descoped and merged at the direction of NASA with a second selected proposal (PI John Baker, PhD; Defining the Relationship Between Simulated Weightlessness and Space Radiation on Cardiovascular Disease, and the Effectiveness of a Countermeasure).” Emerging research demonstrates that low-dose (<0.5 Gy) environmental and medical radiation exposure increases the risk of mortality due to ischemic coronary artery (heart attack) and cerebrovascular (stroke) disease. Results from a recent study of mortality rates among non-flight, low Earth orbit, and Apollo lunar astronauts also suggests, but does not prove, that Apollo lunar astronauts have a higher proportional mortality rate due to cardiovascular disease than non-flight astronauts. Despite the potential for low-dose radiation exposure to have substantial adverse effects on the cardiovascular system, little research has been conducted to investigate this phenomenon. This is particularly urgent as groups of nations and corporate entities are actively planning manned missions back into deep space (i.e., the Moon and Mars), where exposure to galactic comic radiation is pervasive. The present proposal seeks to address some of the fundamental questions that remain outstanding, including: 1) at what dose does simulated space radiation adversely affect cardiac and cerebral vascular function? 2) Does space radiation and weightlessness interact to produce greater coronary and cerebral vascular dysfunction than either alone? 3) Does space radiation and weightlessness have differential effects on coronary and cerebral arteries? 4) Is the cardiovascular dysfunction induced by space radiation and weightless sustained over a long period of recovery, potentially leading to chronic morbidity and mortality? 5) Does dietary intervention (e.g., blackberry supplement) that normally has positive impacts on the cardiovascular system serve as effective countermeasures for radiation-induced cardiovascular dysfunction, and if so, is it effective in both the coronary and cerebral circulations? 6) Is the putative cardiovascular dysfunction induced by space radiation and weightlessness similar in males and females, and is a dietary countermeasure equally effective in both sexes? Results from these studies will lead to a greater understanding of the sustained cardiovascular health risks for prolonged deep space travel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cardiovascular disease risk due to radiation exposure has been changing over the past several decades, with greater understanding that cardiovascular disease risk is elevated with increasingly lower doses of exposure. The current project will add significantly to a very small body of literature on the long-term consequences of low-dose irradiation on the function of cerebral and coronary vascular health, which has a major impact on the risk of stroke and coronary artery disease in humans.
Task Progress:	January 2023 update: Specific Aim 1: All animal work and tissue collection has been completed for Specific Aim 1. Doppler imaging was used to determine general vascular and cardiac function in vivo, and in vitro vascular physiological responses (i.e., endothelium-dependent and endothelium-independent vasodilation) were collected in four sets of vessels, the basilar artery, the coronary artery, the distal internal pudendal artery, and the jugular vein. Doppler and vascular response data analyses are still ongoing. Work to determine vascular structure, vascular mechanics, and protein levels for key markers of nitric oxide signaling and oxidative stress are also ongoing. Doppler pulse wave velocity (PWV) is a measurement of arterial stiffness and is considered an independent predictor of cardiovascular disease risk. Data indicate that simulated space radiation increases vascular stiffness and, correspondingly, may increase long-term risk for cardiovascular disease. Specific Aim 2: Animal work and tissue collection has been partially completed for Specific Aim 2. (Ed., 2/23/23) ---------- Emerging research demonstrates that low-dose (<0.5 Gy) environmental and medical radiation exposure increases the risk of mortality due to ischemic coronary artery (heart attack) and cerebrovascular (stroke) disease. Results from a recent study of mortality rates among non-flight, low-Earth orbit, and Apollo lunar astronauts also suggests, but does not prove, that Apollo lunar astronauts have a higher proportional mortality rate due to cardiovascular disease than non-flight astronauts. Despite the potential for low-dose radiation exposure to have substantial adverse effects on the cardiovascular system, little research has been conducted to investigate this phenomenon. This is particularly urgent as groups of nations and corporate entities are actively planning manned missions back into deep space (i.e., the Moon and Mars), where exposure to galactic comic radiation is pervasive. The present proposal seeks to address some of the fundamental questions that remain outstanding, including: 1) Does simulated space radiation and weightlessness interact to produce greater coronary and cerebral vascular dysfunction than either alone? 2) Does simulated space radiation and weightlessness have differential effects on coronary and cerebral arteries? 3) Is the cardiovascular dysfunction induced by simulated space radiation and weightless sustained over a period of long-term recovery, potentially leading to chronic morbidity and elevated mortality? Results from these studies will lead to a greater understanding of the sustained cardiovascular health risks for prolonged deep space travel.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Willey JS, Britten RA, Blaber E, Tahimic CGT, Chancellor J, Mortreux M, Sanford LD, Kubik AJ, Delp MD, Mao XW. "The individual and combined effects of spaceflight radiation and microgravity on biologic systems and functional outcomes." J Environ Sci Health C Toxicol Carcinog. 2021 Apr 27;39(2):129-79. https://doi.org/10.1080/26896583.2021.1885283 ; PMID: 33902391 , Apr-2021